

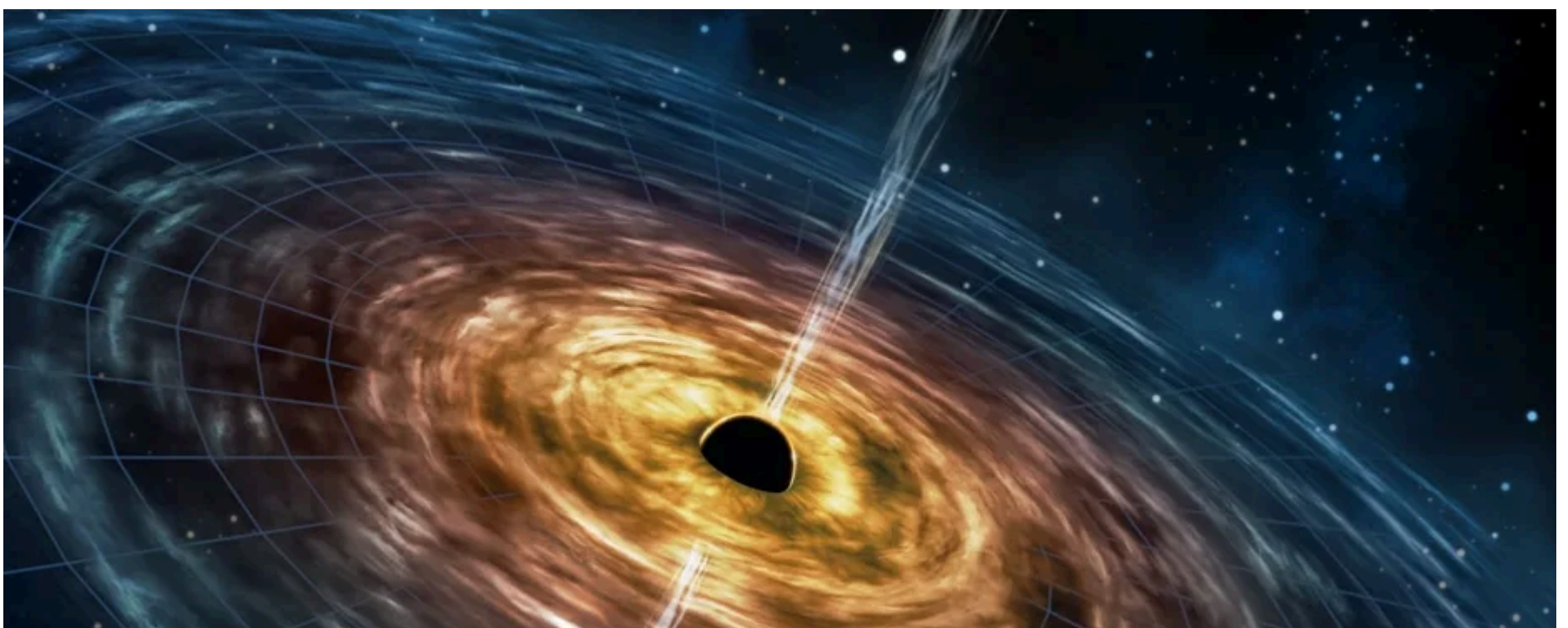
December 19, 2019

Unifying quantum mechanics with Einstein's general relativity

The quantum nature of gravity is an enigma which has eluded even the brightest of [physicists](#) for many decades. Now, Dr Norma G. Sanchez at the French [CNRS LERMA Observatory](#) of Paris-PSL Sorbonne Université describes a possible solution. Her approach takes the form of a general theory, incorporating both quantum mechanics and Einstein's theory of [general relativity](#). If correct, her results could bring researchers a step forward in their knowledge of how the physics which plays out on the very smallest of scales can be compatible with that which occurs on the very largest cosmological scales.

The question of whether light is a wave or a particle has baffled physicists since the 17th century. The problem first emerged when Isaac Newton developed his 'corpuscle' theory, which described particles of light which must only travel in straight lines; as seen in the reflection in a mirror. At the same time, however, other physicists including Christiaan Huygens and Thomas Young proposed that light must propagate as a wave, as it displays properties like diffraction and interference.

In fact, neither of these theories are necessarily 'wrong', since both of them are consistent with experimental observations when the right context is applied. On quantum scales, light travels in packets which we now call photons, but also reliably follows Maxwell's equations, which describe the dynamics of classical electromagnetic waves. To explain how both theories can be correct at the same time, physicists needed a way to unify both classical electrodynamics and quantum physics into a deeper, more all-encompassing theory. Known among physicists as 'quantum electrodynamics', the theory was by no means simple to develop, but nonetheless, it has been done.



appears to behave differently on extremely small scales. In the subsequent decades, the resulting field of quantum mechanics resolved many questions which classical physics didn't have the means to answer. Ultimately, the success of the theory stemmed from the fact that quantum theory is a 'generalised' theory, of which classical physics is just one specialised branch.

This means that while physics appears to be non-classical on quantum scales, classical behaviours emerge on larger scales, where quantum processes become far less relevant overall. Ultimately, therefore, quantum mechanics unites explanations for physical behaviours which are observed on a variety of scales. Yet despite the successes of quantum theory, physicists have realised for some time that even quantum mechanics is not general enough to explain all of the physics we have observed in the universe.

The incompleteness of gravity

Again, first described by Newton, the effects of gravity can be accurately described in many situations using classical physics alone. However, these theories are unable to fully explain all phenomena ever observed by astronomers. Albert Einstein famously resolved this issue through his theory of general relativity. His equations generalised gravity to a more all-encompassing theory; this time, to a geometric model which unites space and time, named spacetime.

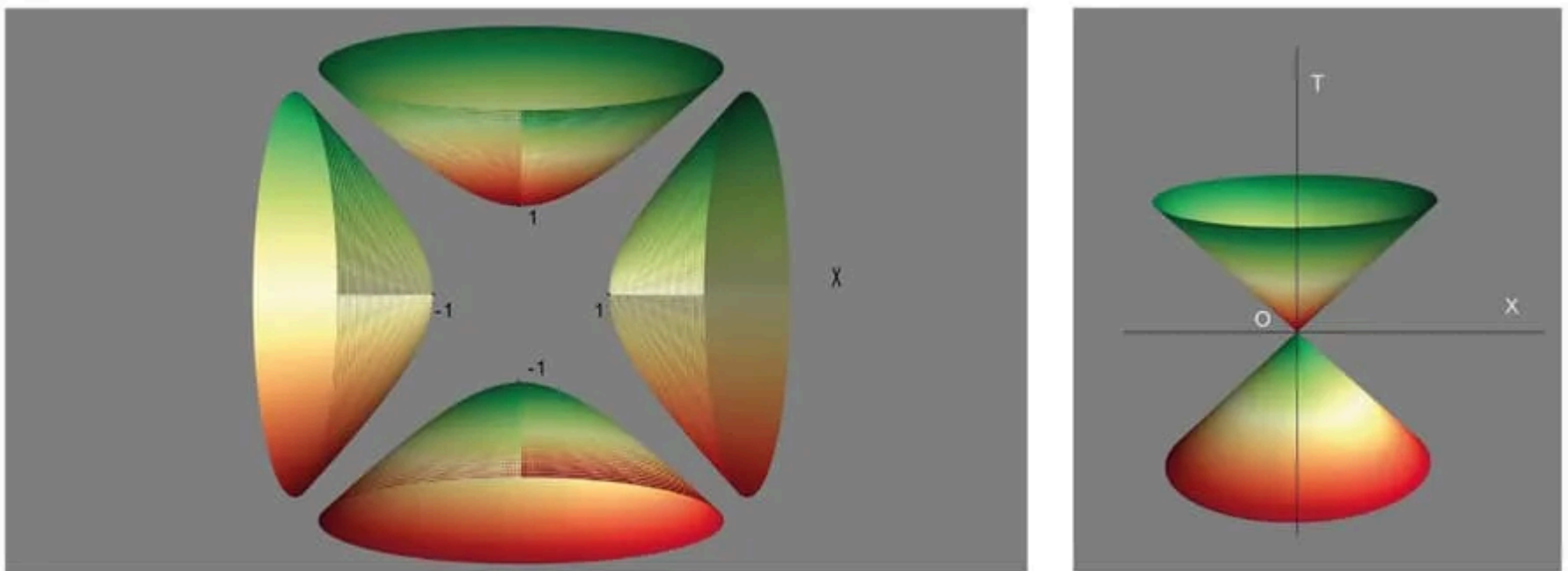


Fig. 1 (left): The quantum light-cone in a space-time diagram (time is the vertical axis). Fig. 2 (right): The known classical light-cone (future and past) of classical relativity in a space-time diagram is a special case of Fig. 1. Copyright Norma Sanchez.

Yet although Einstein's theory has been watertight enough to hold up to even the latest astronomical observations, it appears to be completely incompatible with quantum mechanics. For physicists, this calls for an even deeper generalisation, which has been pre-emptively given the name 'quantum gravity'. In her research, Dr Sanchez aims to realise such a theory, in which both quantum mechanics and general relativity are particular branches which are contained on their appropriate scales. To do this, she has incorporated physical theories which have emerged since Einstein first drew out his famous equations. The key point in her approach is that instead of starting from gravity and quantizing it, she starts from quantum theory and extends it to the high energy scales where gravity and quantum effects are of the same importance.

“ Dr Sanchez uses ‘semi-classical’ gravity as a standard theory, which emerges from her own theories in particular situations. ”

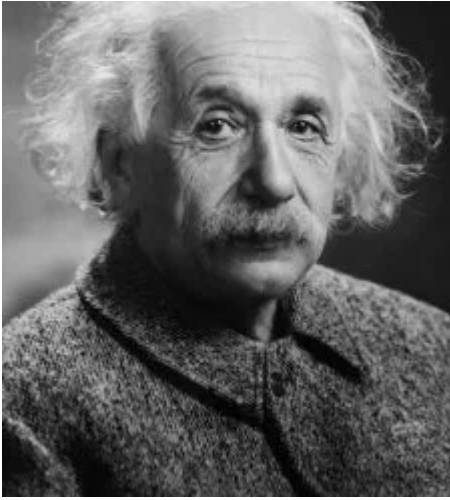
Building on an intermediate theory

Later on in the 20th century, a wide variety of intriguing discoveries about the nature of our universe began to emerge. These included Stephen Hawking's celebrated explanations for how black holes decay through radiation, as well as Cosmic Microwave Background radiation – a faint source of light found across the entire sky, which indicates the density structure of the entire universe.

These theories are united in the approximation that matter displays quantum behaviour, but moves around according to a classical description of spacetime. Known as 'semi-classical' gravity, this model acts as a useful bridge between the separated theories of quantum mechanics and general relativity, and a unified theory of quantum gravity. Because of this, Dr Sanchez uses it as a standard theory, which emerges from her own theories in particular situations.

Extending classical-quantum duality

While previous theories which incorporate semi-classical gravity have described vast astronomical structures, Dr Sanchez brings it down to quantum scales in her research. To describe how gravity could work in this regime, she starts from the classical-



Einstein's theory of general relativity solved many, but not all, issues relating to gravity.

originating from quantum mechanics; Newton's gravitational constant from classical gravitation; and the speed of light, which Einstein proved to be constant through general relativity.

Dr Sanchez has now started a theory which incorporates these three constants into one unified structure. In doing this, the property of 'wave-particle-gravity' duality, or 'classical-quantum gravity' duality emerges. If Dr Sanchez's calculations are correct, it could provide a solution to the puzzle which has eluded physicists for centuries.

Emerging results from generalisation

Within Dr Sanchez's universal theory, wave-particle duality of known quantum physics emerges as a special case for when the right circumstances are applied, instead of being a general rule. This means that instead of being at odds with each other, both theories that light is wave-like and particle-like are simply special branches of the same, more deeply rooted theory.

This also allows for the emergence of properties including the classical-quantum duality of spacetime, whose quantum nature has remained highly elusive so far. In addition, Dr Sanchez claims that an entirely new quantum domain emerges which is not present in the classical description of spacetime – giving physicists a more complete description of its properties. Within this domain, discrete levels of space-time appear at the quantum Planck scale. On classical macroscopic scales, the collective behaviours of these levels appear indistinguishable from the space-time continuum which physicists are more familiar with.



Dr Sanchez's work could transform our understanding of how the universe works. Image credit: Denis Belitsky/Shutterstock.com

Re-drawing the light cone

One particularly important result of Dr Sanchez's theory has been the unexpected emergence of a 'quantum light cone', which doesn't appear within a classical description of the universe. In general relativity, a light cone describes the path taken by light emerging from a single point in time and space; along with all paths taken by the light reaching the point as the flash occurs. The cone shape emerges since all light must reach the same distance from the point at a given time no matter its direction. The boundary of the cone arises because the speed of light is the highest possible velocity throughout the entire universe.

“ One particularly important result of Dr Sanchez's theory has been the unexpected emergence of a 'quantum light cone'. ”

time and space immediately surrounding the flash. On larger scales, where quantum effects are no longer relevant, the classical light cone re-emerges; ultimately allowing both previous theories to work in harmony with each other.

Classical-quantum duality in black holes

Finally, Dr Sanchez has applied her theory of classical-quantum duality to black holes, whose inner workings are masked by an 'event horizon' from which light cannot escape. So far, this has meant that the physical nature of black hole interiors has completely eluded astronomers.



According to Dr Sanchez, the outer regions of black holes are either classical or semi-classical, while the behaviours of their interiors are totally governed by quantum mechanics. The event horizon disappears at the quantum level, a 'quantum border' emerges at which the interiors and exteriors of black holes become the same on quantum scales. Moreover, the classical space-time central singularity – the point at the centre of a black hole at which curvature had been theorised to become infinite, is avoided in this case. According to Dr Sanchez, the singularity must disappear at the quantum level, to remain consistent with the smearing of singularities on quantum scales.

A transformation in understanding

Over the past decades, potential solutions to quantum gravity and several types of mathematical dualities have been hotly debated, and it has remained uncertain whether or not our fundamental understanding of physics needs to be completely rethought. But Dr Sanchez understood that is the physical and universal wave-particle duality of quantum physics which is at the root of the solution. Ultimately, a unified idea of the quantum nature of astronomical-scale phenomena could transform our understanding of how the universe works, and would set the stage for an exciting new era of theoretical physics and cosmology.

Personal Response

What do you find most inspiring about your work?

<itemprop="acceptedAnswer"> The fact that it is a creative, endless work with total freedom, motivated by true advance and discovery beyond current knowledge.

Sanchez, N.G. (2019). [New Quantum Structure of Space-Time](#). *Gravitation and Cosmology*, 25(2), 91-102.

The present article is one from a trilogy published by the author in 2019 on the subject, and from which much more developments are going on: Sanchez, N.G. (2019). [The classical-quantum duality of nature including gravity](#). *International Journal of Modern Physics D28*, No. 03, 1950055.

Behind the Research



Prof Dr Norma G. Sanchez

Prof Dr Norma G. Sanchez is director of research at the French National Centre for Scientific Research and director of the International School Daniel Chalonge – Hector de Vega. Her first work was on the scattering and absorption theory by black holes and Hawking radiation, she obtained for the first time the whole absorption spectrum of the back hole, (French These d'Etat, 1978), connected Einstein equations to other known non-linear theories and their solutions, developed new approaches to quantum fields and strings in curved space-times and found new physical effects. Norma does research in Physical Cosmology, Theoretical Physics, The Standard Model of the Universe from the early to the late universe, with Inflation, the CMB and primordial gravitons, Warm Dark Matter, Dark Energy and quantum physics.

[Contact Details](#)

Research Objectives

Prof Dr Norma Sanchez's theory unifies quantum mechanics with Einstein's general relativity.

Cite this Article

DOI: [10.32907/RO-111-138141](#)

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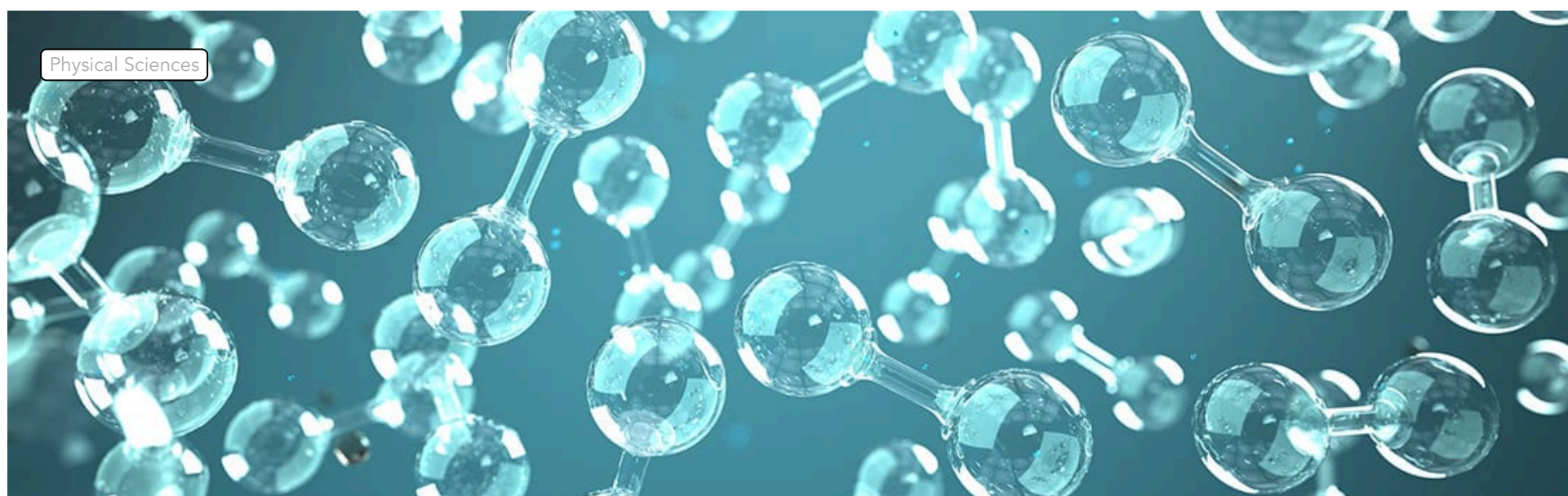
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7 thoughts on “Unifying quantum mechanics with Einstein’s general relativity”

1.  **Richard Waller Jr** says:

[April 8, 2020 at 4:51 pm](#)

If these formulas can be demonstrated this could be the golden goose we have been longing for. We need to keep a close eye on Dr. S.s’ work.

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2.  **Rock Brentwood** says:

[June 18, 2020 at 7:11 pm](#)

I won’t discuss the merits of the idea, but this much needs to be noted and is almost uniformly ignored.

(1) The law of gravity has been verified down to the level of single neutrons <https://physics.aps.org/articles/v11/s34>

(2) Particle physics, which is grounded in Quantum Theory, is well-established

(3) A gravitational monopole solution with the same mass, electroweak and color charge as any of the fundamental particles (possibly except neutrinos) is a NAKED ring singularity – essentially a mini-Stargate. There is serious causality violation in the vicinity of such singularities; even time travel.

Whatever fundamental particles are, if they are not this, then they are something (a) that is not accounted for by General Relativity and (b) lies well within our capacity to problem. If they are this, then (c) we’re on the verge of probing a causality-violating sector of General Relativity, by virtue of (1).

So, any (and all) speculations about the causal structure of spacetime — particularly what that structure looks like from the perspective of Quantum Theory are on the table – including Dr. Sanchez’s.

I believe that the world is a hybrid classical-quantum system; which is a highly non-trivial proposition by virtue of the large number of no-go theorems that populate the minefield; that gravity is classical – the metric, the frame field, the Weyl tensor, the light cone field and gravitational radiation (c.f. “Exclusion of standard $\hbar\omega$ gravitons by LIGO observation” <https://arxiv.org/abs/1712.04437>), though the connection and torsion may be quantized; that the Born Rule itself arises as a consequence of the way in which the classical and quantum sectors interface, particularly under the action of gravitational decoherence.

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3.  **david Meyer** says:

[June 19, 2020 at 9:17 am](#)

A very commendable advance based on dual particle wave behaviour of photons. I have been doing a similar focussed thought process in an amateur way as I am a biochemist who noticed that physics had several large problems. After many floundering years I reach simple conclusions which can explain rather a lot. One may be relevant to Dr Sanchez to progress further, namely that another basic property of photons is that they do not thermodynamically with the universe (hence with universal time) except by gravitation, that is their mass equivalent will alter space and be altered by relativistic spacetime without their wave function collapsing, hence they do cause slight movements of the macroscopic world and therefore slightly affect the thermodynamic evolution of the universe (time) whilst remaining quantised. In other interactions time only passes (for the universe) when the ‘wave function collapses’ and the energy is absorbed. The quantised world is therefore a different time zone / dimension / or if the end of the universe is a dispersed photon cloud, it all takes place under conditions of time at the end of the universe.

(My own conclusions agree that it is time and scale that separate the main physical theories, but that time itself should be represented as universal time or more specifically Universal dates, which actually is entropy. This could be evaluated in physical formulae I think simply by placing a minimal S (entropy) for the start of the universe and a maximal S for the end – this could be 0 and 1 when normalised, and using it instead of ‘t’ (seconds). Then all quantised materials would have an S

red-shift, maybe it is not the result of space expanding, but the cause, ie. light is dark energy, could it be named any worse? and the energy lost increases the wavelength of the light.)

I wanted to try and use the dimensions of entropy inserted for time in the equations of relativity and quantum theory to see what sense might arise, but I do not have the mathematical ability. Maybe my ideas will reach someone who has.

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4.  **Silvio** says:

[July 26, 2020 at 6:14 pm](#)

Spyroe theory uses similar ideology but shows the light cones ends not as a hyperbola but a single helix which connects to the outbound light cone.

Doing this on the 3 axis forms a shape that can represent one quantum of energy, displaying characteristics of both general relativity and quantum mechanics.

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5.  **Caleigh Fisher** says:

[September 6, 2020 at 8:22 pm](#)

Have you used this using a 5d manifold?

With a degenerate Toroid one realizes that the inside of event horizon shares the same surface as the outside of everywhere. All geodesic end up at event horizon which is concentration of mass/energy since the flow is always downhill. Towards density.

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6.  **ian bartlett** says:

[March 30, 2021 at 1:38 pm](#)

Possibly/probably a stupid question; But what existed before the Big Bang?

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7.  **James Rey** says:

[August 7, 2021 at 4:48 pm](#)

I call it quantum relativity

More or less if time is not linear than you can combine both theories

And I believe what we think is time is not

Everything had happened before and we just can't completely see it

Free will is still exists but how can predestination exist

Well if you say time is not linear and everything happened then it's not predestination it's destination

I have been writing a book and I finally came to that conclusion

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